



ANALOG ELETRONICS

LECTURE 4:

FIELD-EFFECT TRANSISTORS (FETS)

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OUTLINE

- Introduction
- Types of FETs
- The JFETs
- Difference between JFET and BJT
- The MOSFET
- Types of MOSFETs



INTRODUCTION

- In BJT, both **holes** and **electrons** play part in the conduction process thus the term **bipolar** transistor.
 - The BJT has two principal disadvantages,
 - **Low input impedance** because of forward biased emitter junction and
 - Has **considerable noise** level.
 - The field effect transistor (FET) has, **large input impedance** which may be more than 100 M Ω .
 - The FET is generally much **less noisy** than the BJT.
 - In this lecture, we shall focus our attention on the **construction, working** and **circuit applications** of field effect transistors.
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TYPES OF FIELD EFFECT TRANSISTORS

There are two basic types of field effect transistors;

- Junction field effect transistor (JFET)
- Metal oxide semiconductor field effect transistor (MOSFET)



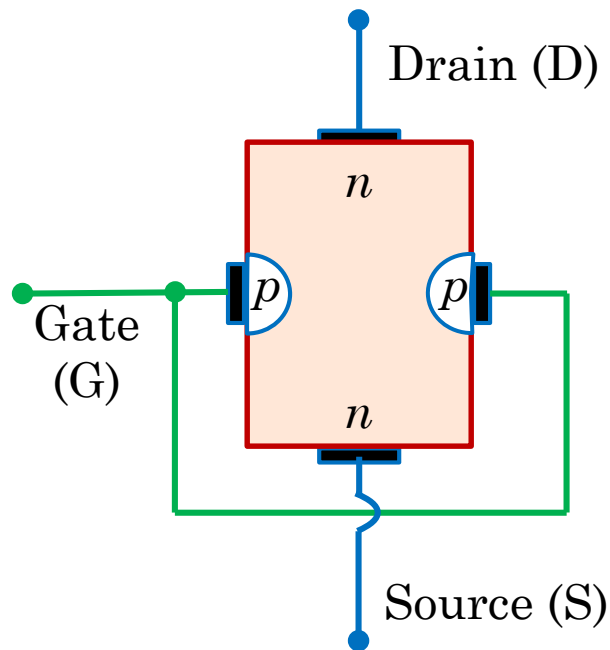
JUNCTION FIELD EFFECT TRANSISTOR (JFET)

- A junction field effect transistor is a three terminal semiconductor device in which current conduction is by one type of carrier i.e., **electrons** or **holes**.
- The JFET has **high input impedance** and **low noise level**.
- The current conduction is controlled by means of an electric field.

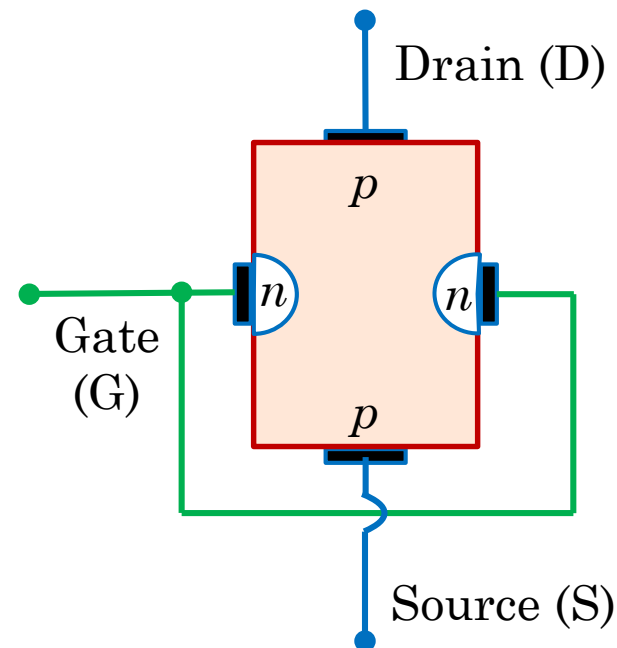


CONSTRUCTION

- A JFET consists of a **p-type** or **n-type** silicon bar containing **two pn junctions** at the sides.
- The bar forms the **conducting channel** for the charge carriers.
- If the bar is of n-type, it is called **n-channel JFET**.
- If the bar is of p-type, it is called a **p-channel JFET**.



n-channel JFET



p-channel JFET

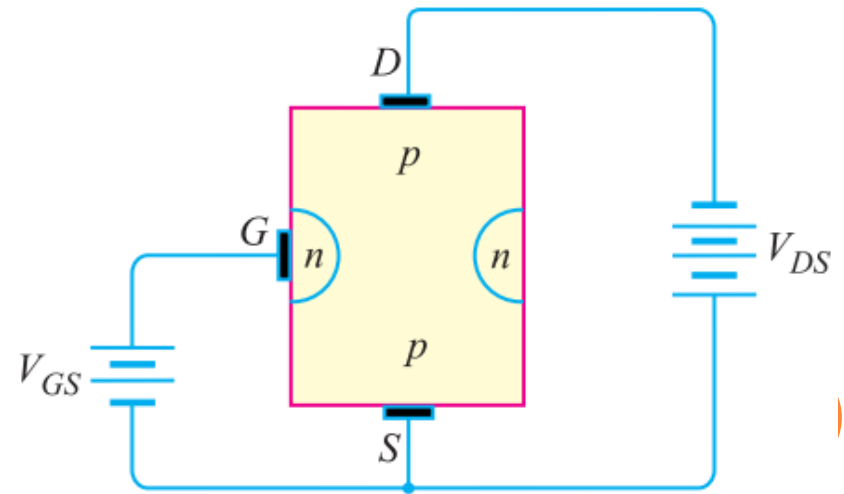
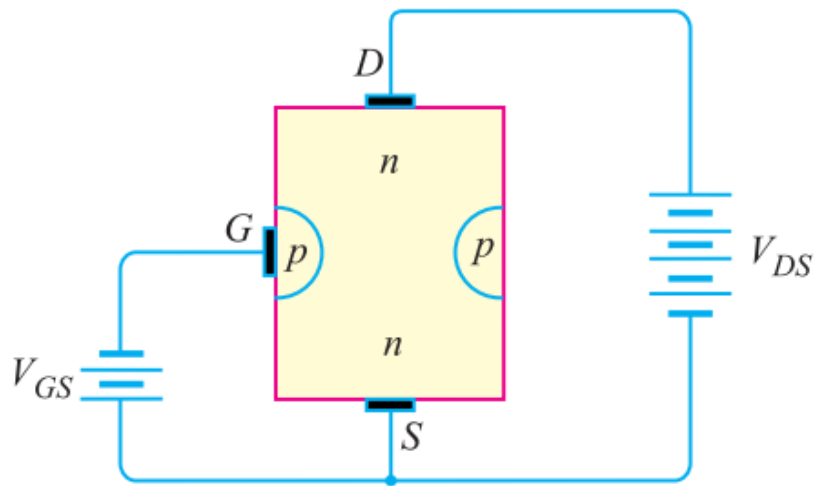
CONSTRUCTION

- The two pn junctions forming diodes are connected internally and a common terminal called **gate** is taken out.
- Other terminals are **source** and **drain** taken out from the bar.
- Thus a JFET has essentially three terminals viz., gate(G), source (S) and drain (D).



JFET POLARITIES

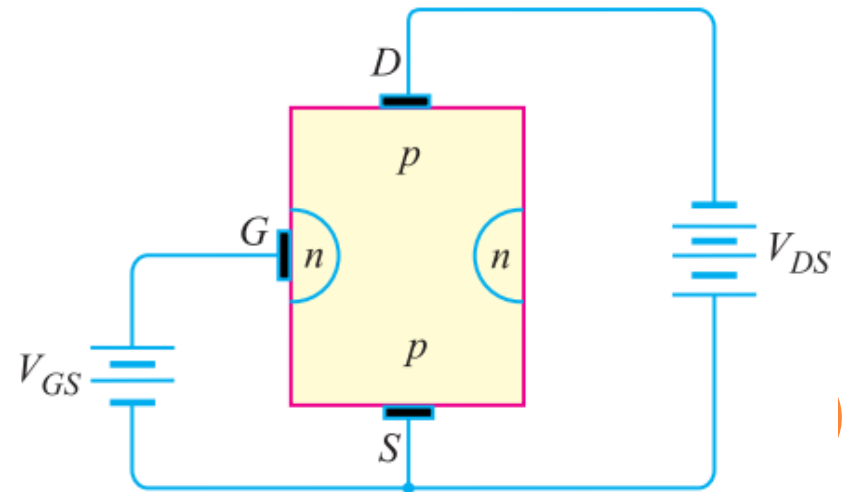
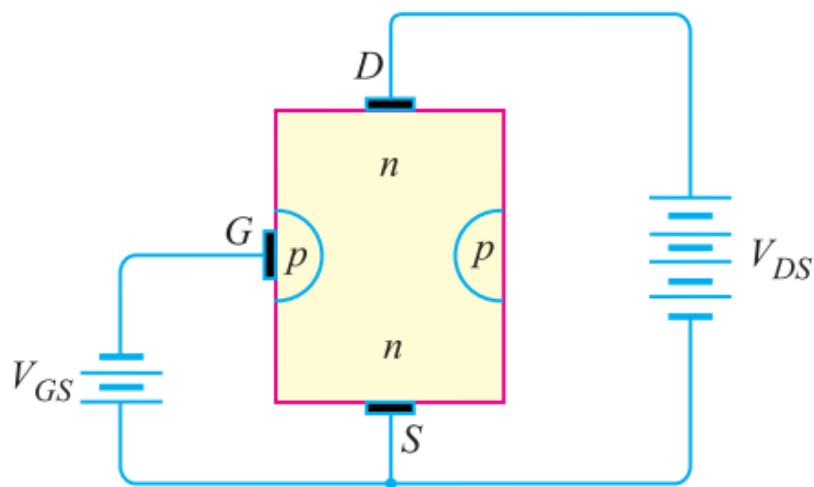
- The voltage between the gate and source is such that the gate is reverse biased.
- This is the normal way of JFET connection.
- The drain and source terminals are interchangeable i.e., either end can be used as source and the other end as drain.



JFET POLARITIES

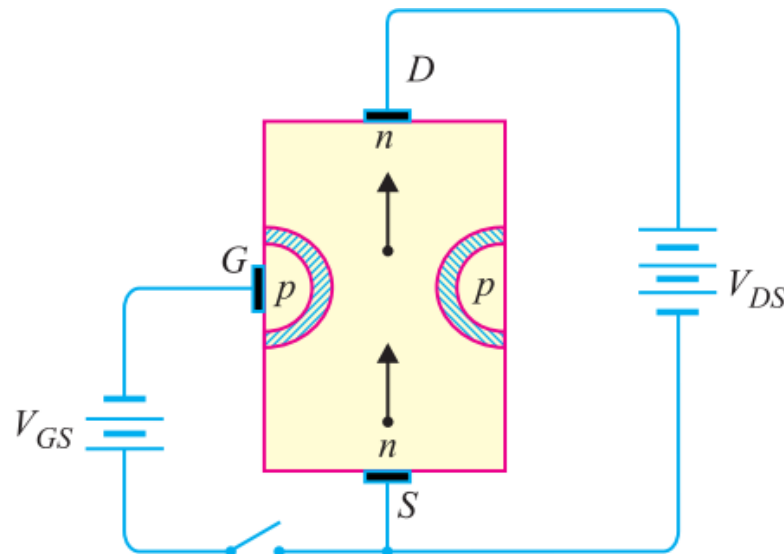
The following points may be noted :

- (i) The input circuit (i.e. Gate to source) of a JFET is reverse biased. This means that the device has high input impedance.
- (ii) The drain is so biased w.r.t. source that drain current I_D flows from the source to drain.
- (iii) In all JFETs, source current I_S is equal to the drain current i.e. $I_S = I_D$



PRINCIPLE OF JFET

- The **two pn junctions** at the sides form **two depletion layers**.
- The current conduction by **charge carriers** is through the channel between the two depletion layers and out of the drain.
- The **width** and hence **resistance** of this channel can be controlled by changing the input voltage V_{GS} .



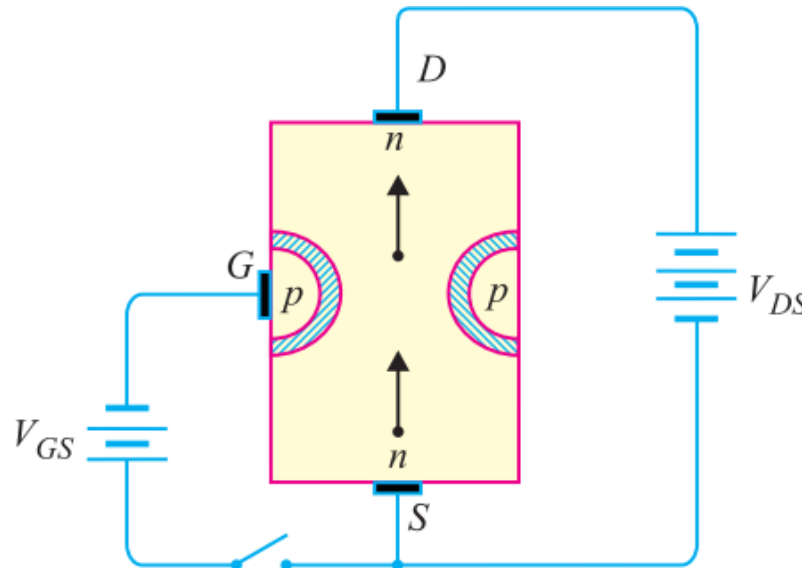
PRINCIPLE OF JFET

- The greater the reverse voltage V_{GS} , the wider will be the depletion layers and narrower will be the conducting channel.
- The narrower channel means greater resistance and hence source to drain current decreases.
- Thus JFET operates on the principle that width and hence resistance of the conducting channel can be varied by changing the reverse voltage V_{GS} .
- In other words, the magnitude of drain current (I_D) can be changed by altering V_{GS} .



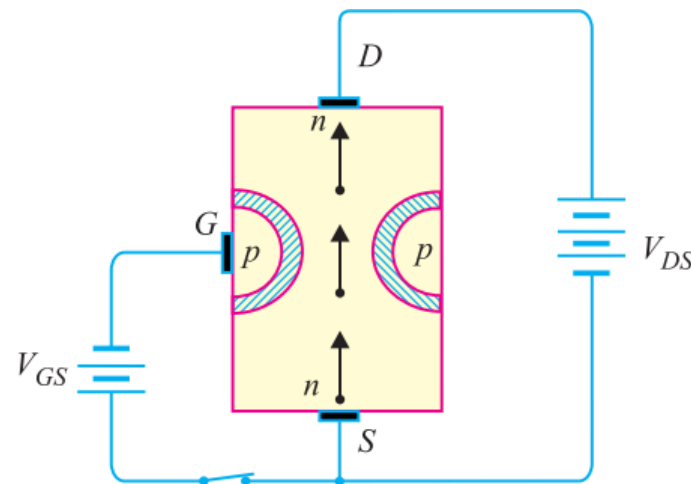
WORKING OF JFET

- When a voltage V_{DS} is applied between drain and source terminals and voltage on the gate is zero the two pn junctions at the sides of the bar establish depletion layers.
- The electrons will flow from source to drain through a channel between the depletion layers.
- The size of these layers determines the width of the channel and hence the current conduction through the bar.



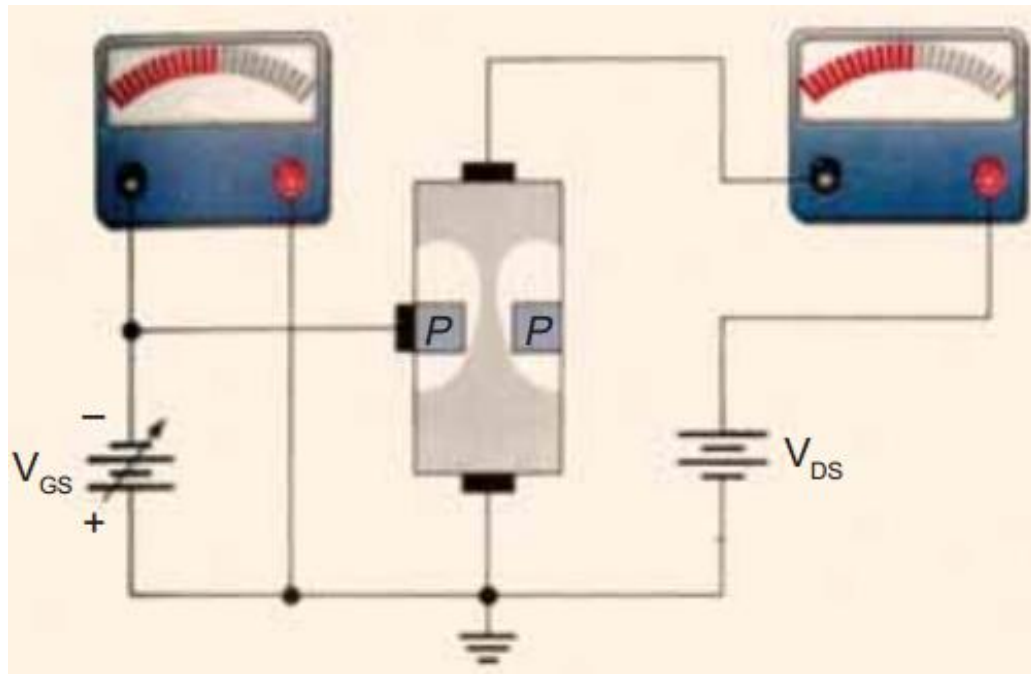
WORKING OF JFET

- When a reverse voltage V_{GS} is applied between the gate and source, the width of the depletion layers is increased.
- This reduces the width of conducting channel, thereby increasing the resistance of n-type bar.
- Therefore, the current from source to drain is decreased.
- If voltage V_{GS} is decreased, the width of the depletion layers also decreases, thus increases the width of the conducting channel and hence source to drain current.

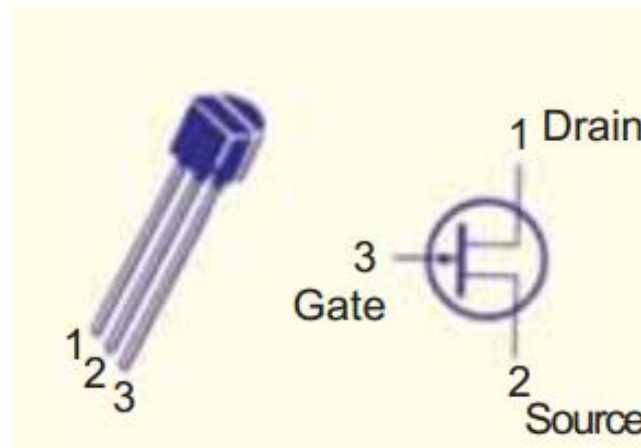
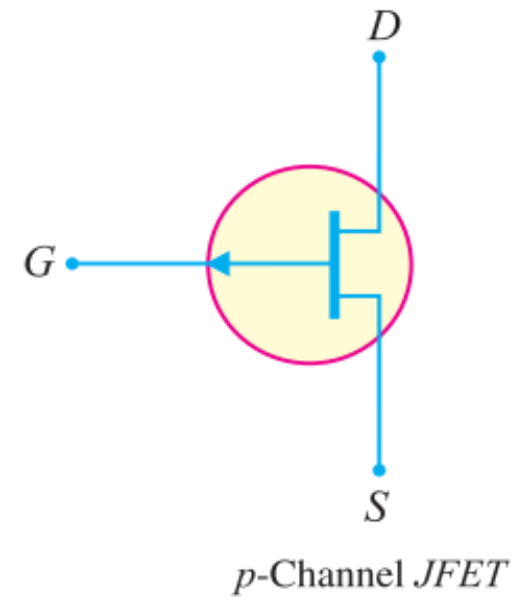
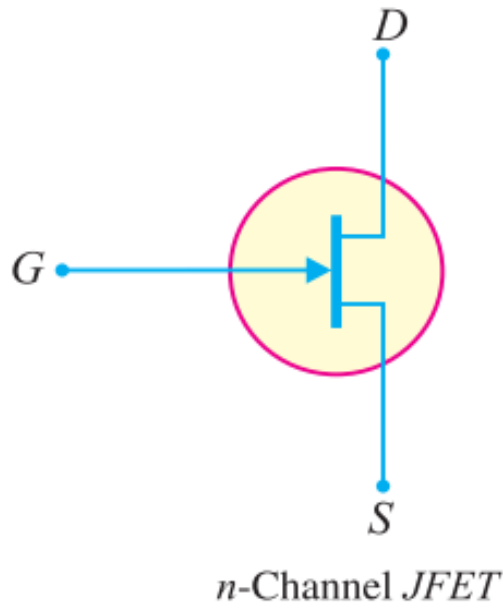


WORKING OF JFET

- The current from source to drain is controlled by the application of potential (i.e. electric field) on the gate.
- For this reason, the device is called **field effect transistor**

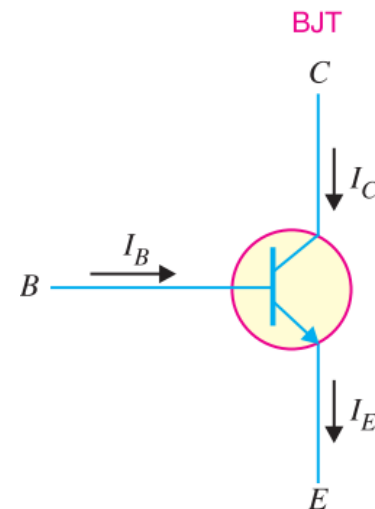
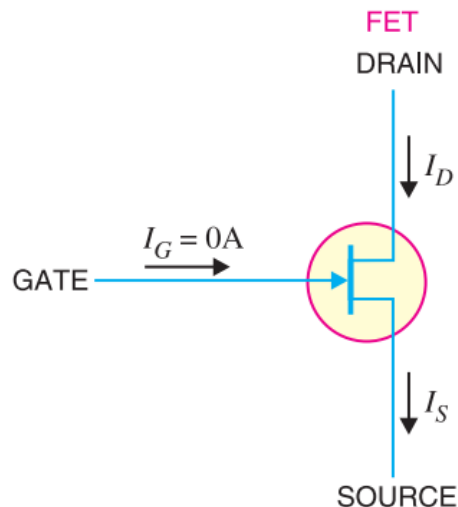


SCHEMATIC SYMBOL OF JFET



DIFFERENCE BETWEEN JFET AND BJT

- JFET is a unipolar transistor while BJT is a bipolar transistor.
- No current (actually, a very, very small current) enters the gate of JFET(i.e. $I_G = 0A$) while typical BJT base current might be a few μA .



DIFFERENCE BETWEEN JFET AND BJT

- JFET input circuit is reverse biased, hence **high input impedance**, while BJT input circuit is forward bias hence **low input impedance**.
- BJT is **current** controlled device while whereas JFET is a **voltage** controlled device.
- In JFET, there are **no junctions** as in BJT. The conduction is through an n- type or p-type semiconductor material hence noise level is very small.



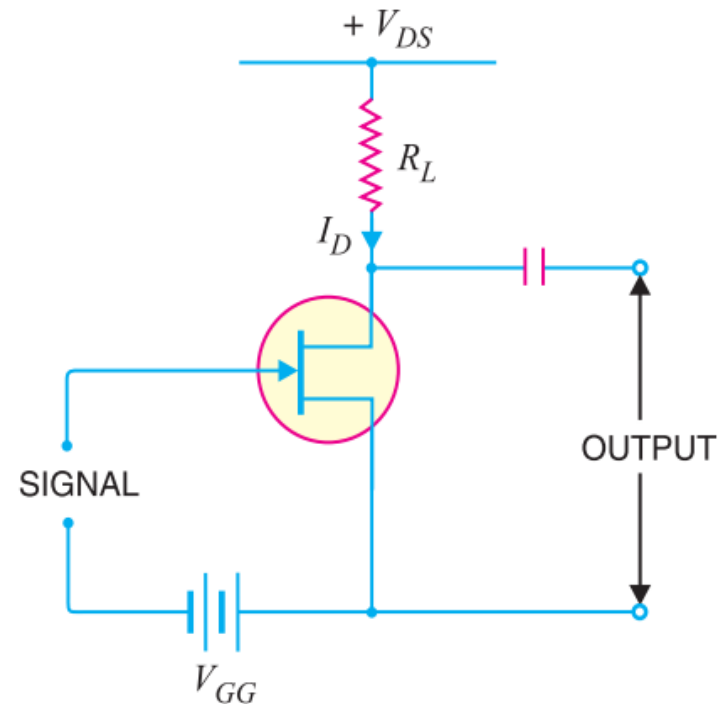
JFET AS AN AMPLIFIER

- The weak signal is applied between gate and source and amplified output is obtained in the drain-source circuit.
- For the proper operation of JFET, the gate must be negative w.r.t. source i.e., input circuit should always be reverse biased.
- This is achieved either by inserting a battery V_{GG} in the gate circuit or by a circuit known as biasing circuit.
- In the present case, we are providing biasing by the battery V_{GG} .



JFET AS AN AMPLIFIER

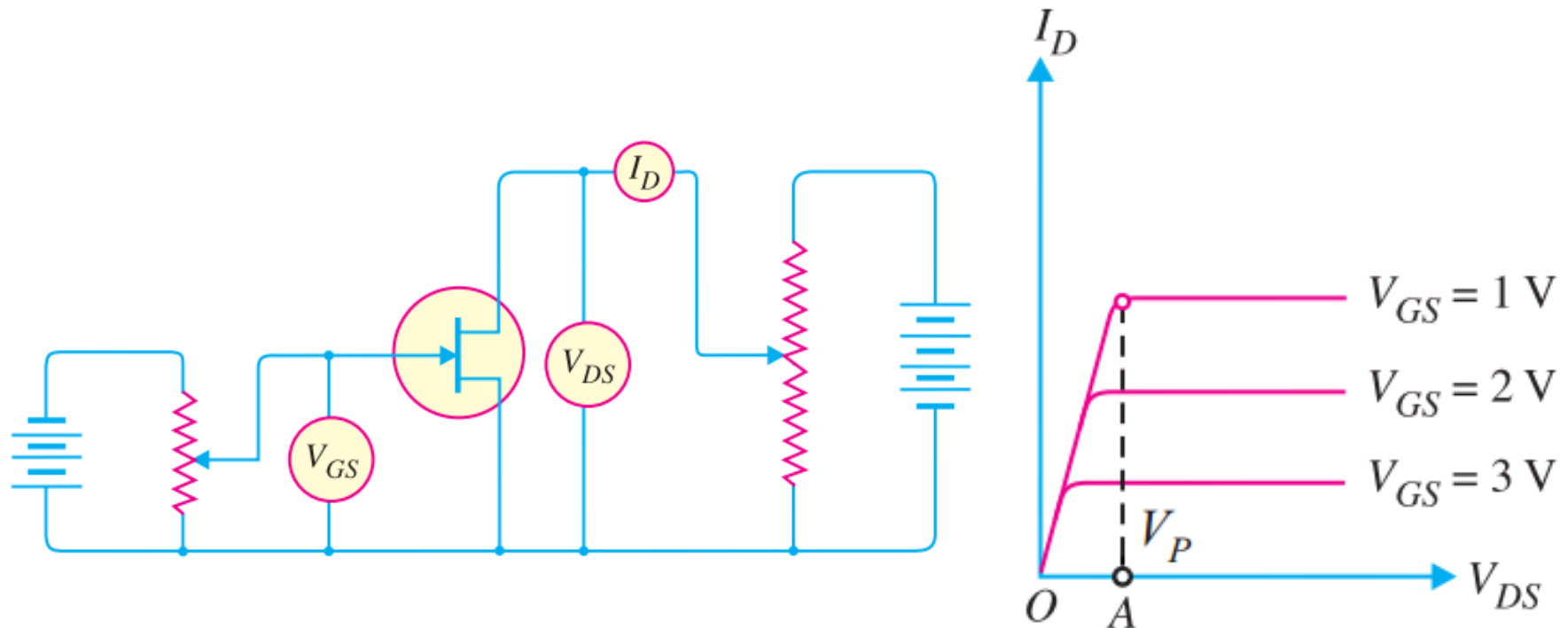
- A small change in the reverse bias on the gate produces a large change in drain current.
- This fact makes JFET capable of raising the strength of a weak signal.
- During the positive half of signal, the reverse bias on the gate decreases.
- This increases the channel width and hence the drain current.
- During the negative half-cycle of the signal, the reverse voltage on the gate increases.
- Consequently, the drain current decreases.
- The result is that a small change in voltage at the gate produces a large change in drain current.



- These large variations in drain current produce large output across the load R_L
- In this way, JFET acts as an amplifier

OUTPUT CHARACTERISTICS OF JFET

- The curve between drain current (I_D) and drain-source voltage (V_{DS}) of a JFET at constant gate source voltage (V_{GS}) is known as output characteristics of JFET.



SALIENT FEATURES OF JFET

- A JFET is a three-terminal voltage-controlled semiconductor device i.e. input voltage controls the output characteristics of JFET.
- The JFET is always operated with gate-source pn junction reverse biased.
- The gate current is zero i.e. $I_G = 0\text{ A}$, thus $I_D = I_S$
- The JFET must be operated between V_{GS} and $V_{GS}(off)$
- For this range of gate-to-source voltages, I_D will vary from a maximum of I_D to a minimum of almost zero.
- Because the two gates are at the same potential, both depletion layers widen or narrow down by an equal amount.
- The JFET is not subjected to thermal runaway when the temperature of the device increases.
- The drain current I_D is controlled by changing the channel width.
- Since JFET has no gate current, there is no β rating of the device.

ADVANTAGES OF JFET

- It has a **very high input impedance**. This permits high degree of isolation between the input and output circuits.
- **Less noise** compared to BJT.
- A JFET has a **negative temperature co-efficient of resistance**. This avoids the risk of thermal runaway.
- A JFET has a **very high power gain**. This eliminates the necessity of using driver stages.
- A JFET has a **smaller size, longer life and high efficiency**.



METAL OXIDE SEMICONDUCTOR FET (MOSFET)

- The main drawback of JFET is that **its gate** must be **reverse biased** for proper operation of the device.
- This means that we can only decrease the width of the channel (i.e. decrease the conductivity of the channel) from its zero-bias size.
- This type of operation is referred to as **depletion-mode** operation.
- Therefore, a JFET can only be operated in the depletion-mode.

METAL OXIDE SEMICONDUCTOR FET (MOSFET)

- However, there is a field effect transistor (FET) that can be operated to enhance (or increase) the width of the channel.
- With consequent increase in conductivity of the channel.
- i.e.it can have enhancement-mode operation.
- Such a FET is called MOSFET.



METAL OXIDE SEMICONDUCTOR FET (MOSFET)

- A field effect transistor (FET) that can be operated in the enhancement-mode is called a MOSFET.
- A MOSFET is an important semiconductor device and can be used in any of the circuits covered for JFET.
- However, a MOSFET has several advantages over JFET including **high input impedance** and **low cost of production**.



TYPES OF MOSFETs

1. Depletion-type MOSFET or D-MOSFET

- The D-MOSFET can be operated in both the depletion-mode and the enhancement-mode.
- For this reason, a D-MOSFET is sometimes called depletion/enhancement MOSFET.

2. Enhancement-type MOSFET or E-MOSFET.

- The E-MOSFET can be operated only in enhancement-mode.

